

NEGF Study of Room-Temperature IETS in a BEOL-Compatible ZnO/Mg_{0.3}Zn_{0.7}O RTD for a Quantum Biomimetic Electronic Nose

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Abstract—We present a quasi-3D Non-Equilibrium Green’s Function (NEGF) study of a back-end-of-line (BEOL) compatible ZnO/Mg_{0.3}Zn_{0.7}O resonant tunneling diode (RTD) operated as a room-temperature electronic nose using Inelastic Electron Tunneling Spectroscopy (IETS). A projected self-consistent Born approximation (SCBA) is derived for a localized molecular vibron whose transverse-mode coupling matrix is rank-1 (up to corrections $O((\sigma_{\text{mol}}k_{\perp})^2) \sim 10^{-6}$ for realistic molecular sizes $\sigma_{\text{mol}} \approx 3 \text{ \AA}$). This reduces the full 3D phonon NEGF problem to a single projected 1D SCBA loop.

For a device of cross-section $10 \mu\text{m} \times 10 \mu\text{m}$ at 10–300 K, Anderson-mixing (DIIS) converged SCBA results in resolvable molecule-specific d^2I/dV^2 signatures for synthetic single and two-mode analytes (100 meV and 180 meV modes). This enables practical room-temperature IETS fingerprinting without cryogenic cooling, leveraging the RTD resonance as an energy filter.

I. INTRODUCTION

Inelastic Electron Tunneling Spectroscopy (IETS) provides vibrational fingerprints but traditionally requires cryogenic temperatures [1]. Patil [2] demonstrated that a double-barrier RTD can serve as an intrinsic narrow-band energy filter, preserving d^2I/dV^2 features at 300 K using a toy device. We extend this concept to an all-oxide stack: ZnO/Mg_{0.3}Zn_{0.7}O, the only published BEOL-compatible oxide pair in the $\Delta E_c \approx 0.5 \text{ eV}$ window exhibiting NDR [3] [4]. We quantitatively study its ability to discriminate molecular vibrational spectra at room temperature using a rigorously derived *rank-1 projected NEGF-SCBA*.

For a Gaussian molecular approximation ($\sigma_{\text{mol}} = 3 \text{ \AA}$), the electron-vibron coupling factorizes as $M = D_0\chi(z)|u\rangle\langle u|$ in the transverse-mode basis (correction $O(10^{-6})$). This reduces the 3D SCBA to a scalar 1D projected Dyson equation, exact in the continuum limit.

II. DEVICE AND METHOD

Device. Symmetric RTD: 2 nm Mg_{0.3}Zn_{0.7}O barriers, 3 nm ZnO well, 10 nm n^+ -ZnO contacts ($N_D = 10^{25} \text{ m}^{-3}$). Parameters: $\Delta E_c = 0.47 \text{ eV}$, $m^* = 0.28m_0$, $\hbar\omega_{\text{LO}} \approx 72 \text{ meV}$. NDR occurs near $V_{\text{res}} \approx 560 \text{ mV}$ with peak current $\sim 69 \text{ nA}$ on a $10 \mu\text{m} \times 10 \mu\text{m}$ footprint (see Fig. 1).

NEGF + rank-1 SCBA. Longitudinal discretization uses a 0.2 nm tight-binding grid. The retarded Green’s function follows the standard form with semi-infinite leads [5], symmetric bias ($\mu_{L,R} = E_F \pm V/2$, $E_F = 20 \text{ meV}$), and causal phonon self-energy. Bulk LO phonons are mode-diagonal; molecular

vibrons add a rank-1 perturbation. The combined self-energy remains diagonal + rank-1, reducing to additive 1D SCBA (see Fig. 2).

Bulk + molecular phonons. The bulk ZnO LO phonon ($\hbar\omega_{\text{LO}} = 72 \text{ meV}$) couples uniformly in the transverse plane. It is *always* present ($D_{\text{bulk}}^2 = 0.001 \text{ eV}^2/\text{site}$, $\chi \equiv 1$ in the active region. Molecular vibrons are added on top with $D_0^2 = 0.1 \text{ eV}^2$ and Gaussian $\chi(z)$ centered on the emitter barrier. The combined self-energy $\Sigma = \Sigma^{\text{bulk}} + \Sigma^{\text{mol}}$ is diagonal + rank-1.

Convergence. Anderson mixing [6] (DIIS, history $M = 8$, $\beta = 0.3$) ensures robust convergence (10–60 iterations, current conservation $< 0.2\%$).

Analytes. Three synthetic molecules: Mol_A ($\hbar\omega_A = 100 \text{ meV}$), Mol_B ($\hbar\omega_B = 180 \text{ meV}$), Mol_AB (both), and Baseline with only the bulk 72 meV phonon. These energies lie in the vibrational fingerprint band (50–200 meV) [7].

III. RESULTS AND DISCUSSION

All analytes show clean NDR at $\sim 560 \text{ mV}$ (Fig. 3a). Molecular modes produce distinct ΔI steps in the pre-NDR region and clear peaks in d^2I/dV^2 (Fig. 3b). The discrimination metric $\Delta D(V) = d^2I_{\text{mol}}/dV^2 - d^2I_{\text{BL}}/dV^2$ isolates vibrational signatures (Fig. 3c), with Mol_AB showing near-linear superposition due to rank-1 structure. Temperature dependence (10–300 K, Fig. 4) confirms that IETS features remain resolvable at 300 K thanks to the RTD’s energy filtering.

In conclusion, we show that a BEOL-compatible oxide RTD can enable room-temperature molecular discrimination via IETS. The computational approach used also provides an efficient path for phonon-mediated transport in a 3D stack with rank-1 projected NEGF-SCBA. Future work includes self-consistent Poisson, DFT-calculated couplings, and real VOC spectra [7].

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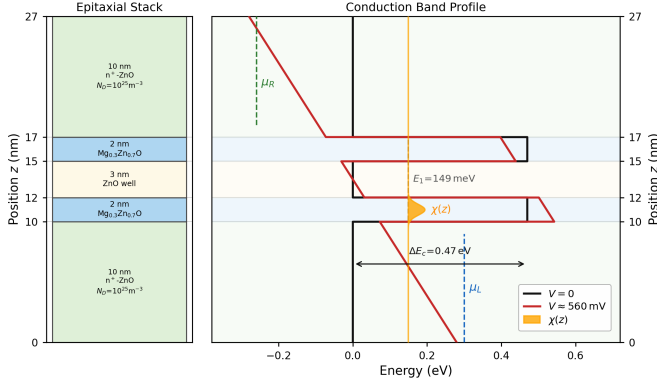


Fig. 1: Schematic structure (left) and conduction-band profile (right) of the symmetric ZnO/Mg_{0.3}Zn_{0.7}O RTD (2/3/2 nm active region). (i) $V = 0$: bound state $E_1 = 149$ meV; orange Gaussian $\chi(z)$ marks molecular coupling. (ii) At resonance ($V \approx 560$ mV): E_1 aligns with μ_L . Symmetric bias: $\mu_{L,R} = E_F \pm V/2$.

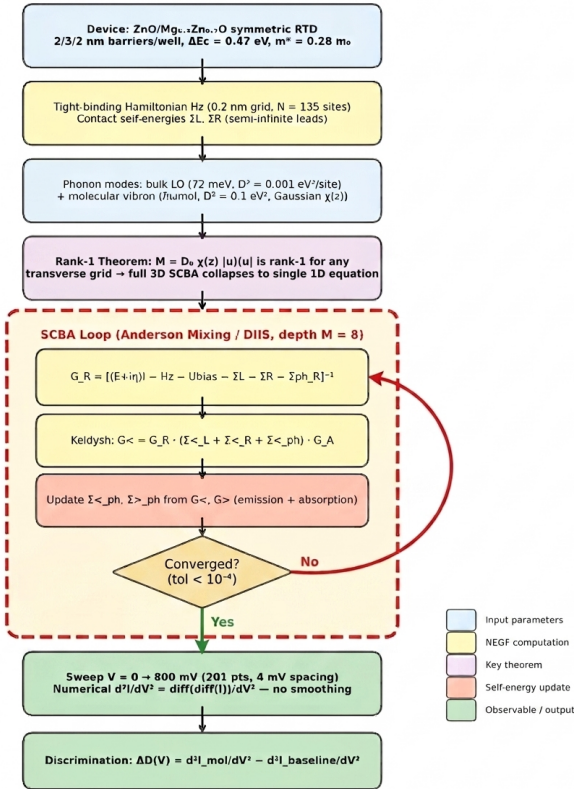
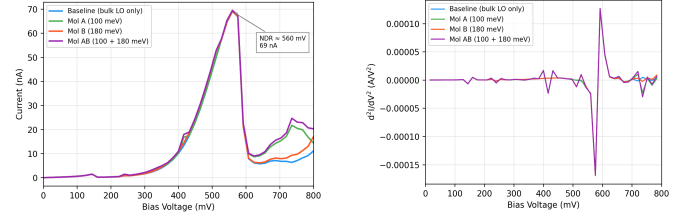


Fig. 2: Rank-1 projected NEGF-SCBA methodology. Molecular coupling is rank-1 in transverse modes for any grid size, reducing 3D SCBA to one 1D projected equation. Anderson mixing (DIIS, $M = 8$, $\beta = 0.3$) ensures convergence with current conservation $< 0.2\%$. d^2I/dV^2 computed by numerical double-differentiation of converged $I(V)$.

TABLE I: Candidate RTD material stacks for room-temperature IETS. A viable sensor requires CBO ~ 0.5 eV (NDR voltage in the 50–400 meV vibrational band), demonstrated NDR, and BEOL compatibility ($< 400^\circ\text{C}$ deposition). Only ZnO/Mg_{0.3}Zn_{0.7}O satisfies all criteria.

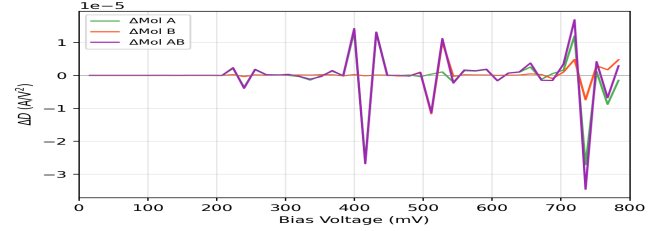
Material stack	ΔE_c (eV)	m^*/m_0	NDR	BEOL	Ref.
AlGaAs/GaAs	0.23	0.067	Yes	No	[8]
GaN/AlN	1.8	0.20	Yes	No	[9]
In ₂ O ₃ /κ-Ga ₂ O ₃	0.45	0.30	No	Yes	[10] [11]
ZnO/Mg_{0.3}Zn_{0.7}O	0.47	0.28	Yes	Yes	[3] [‡] [4]

[‡] Krishnamoorthy *et al.* used $x = 0.2$; this simulation uses $x = 0.3$ (CBO scaling $1.57x$ [12]).



(a) I-V at 300 K.

(b) d^2I/dV^2 at 300 K.



(c) Discrimination metric $\Delta D(V)$ at 300 K.

Fig. 3: Simulation results at 300 K for Baseline, Mol_A ($\hbar\omega = 100$ meV), Mol_B ($\hbar\omega = 180$ meV), and Mol_AB on a $10 \mu\text{m} \times 10 \mu\text{m}$ ZnO/Mg_{0.3}Zn_{0.7}O RTD (51 bias points, 16 mV spacing). (a) I-V: NDR at ~ 560 mV, Baseline peak ~ 69 nA (PVR ≈ 6). (b) Numerical d^2I/dV^2 : molecular modes produce species-specific deviations (c) Discrimination metric $\Delta D = d^2I_{\text{mol}}/dV^2 - d^2I_{\text{BL}}/dV^2$. Mol_AB is approximately the superposition of Mol_A and Mol_B corrections, consistent with rank-1 additive structure.

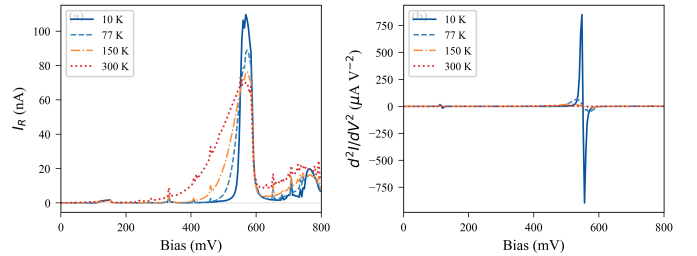


Fig. 4: Temperature dependence (10–300 K) for Mol_A ($\hbar\omega = 100$ meV). (a) I-V: peak current rises from ~ 69 nA (300 K) to ~ 110 nA (10 K) as Fermi smearing decreases; NDR sharpens. (b) d^2I/dV^2 : the 100 meV IETS peak narrows at low T but remains clearly resolved at 300 K, confirming RTD energy filtering preserves IETS features beyond the thermal broadening limit $3k_B T$.